



Microstructure and mechanical properties in laser transmission bonding of 5052AA/glass joints using nanosecond pulsed laser



Hoai Nguyen^a  , Jeng-Rong Ho^b

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<https://doi.org/10.1016/j.ceramint.2025.07.195> 

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Abstract

In this study, we used a nanosecond pulsed laser for laser transmission bonding to examine the heterogeneous bonding mechanism of glass and aluminum alloy. Our study concentrated on investigating different processing parameters as well as the impact of pulse overlapping factors and single-pulse energy on bonding quality. A thorough examination using SEM and EDS revealed that during welding, Al and SiO₂ experienced intra-mixing and inter-particle diffusion in the molten pool. The work comprehensively determines the weld creation mechanism by carefully examining elemental compositions, phase forms, and cross-sectional structures. At the weld interface, polycrystalline aluminum alloy nanoparticles, aluminum alloy oxides, and an Al-O-Si complex were detected by HR-TEM and XPS measurements. The strong bond between glass and aluminum alloy was further enhanced by these new substances. We used shear-tensile separation force measurements to assess weld strength. The highest

bond strength over 50.08MPa was attained, exceeding previously documented results with a single-pulse energy of 0.541 mJ and a pulse overlapping factor of 90%. This work emphasises how the energy of a single pulse directly correlates with the keyhole zone's growth, whereas the molten zone's heterogeneity is determined by the pulse overlapping ratio. This study shows that it is possible to create high-quality glass/aluminum alloy joints using an affordable laser source, which holds promise for the production of heat-dissipation substrates for semiconductor devices.



Keywords

Laser transmission bonding; Glass-aluminum alloy bonding; Modified keyhole zone; Single-pulse energy; Bonding shear strength

1. Introduction

Glass has long been an important material due to its numerous beneficial properties. These properties include a low coefficient of thermal expansion, high electrical resistivity, high stiffness, strong corrosion resistance, and low dielectric constant. As a result, glass has been a versatile and innovative material, playing a crucial role in the development of various products, such as microelectromechanical systems (MEMS) [1], photonics [2], microfluidics [3], optoelectronics [4], and optical telecommunications [5]. On the other hand, aluminum alloy offers advantages like low density, minimal sensitivity to magnetic fields, and high electrical and thermal conductivity [6]. Therefore, combining glass and aluminum alloy is essential to fully utilize the benefits of both materials.

Over the past few decades, various effective methods for bonding dissimilar materials have emerged, including mechanical joining [7], ultrasonic bonding [8], induction welding [9], laser welding [10,11], and diffusion bonding [12]. Among these techniques, laser welding has gained significant attention due to its numerous advantages, such as non-contact operation, high flexibility, exceptional precision, and environmental friendliness. One notable benefit of laser welding is its ability to perform

internal pattern welding through a process known as laser transmission bonding (LTB). In this method, the laser beam passes through the upper glass and interacts with the metal located beneath it, facilitating a strong and efficient bond.

Ultrashort pulse lasers, such as femtosecond (fs) and picosecond (ps) pulsed lasers, have recently gained significant attention for bonding applications due to their remarkable precision in delivering welding energy within a small and well-defined range. These lasers produce a minimal heat-affected zone, making them ideal for precise and effective bonding. For instance, Tan et al. [13] investigated the relationship between single-pulse energy and linear energy density in ps pulsed laser-welded glass/rough 6061 Al samples, achieving a maximum bonding strength of 15.98MPa. Huo et al. [14] utilized an fs pulsed laser to weld rough copper to glass, reaching a shear strength of approximately 17.19MPa. Additionally, Shen et al. [15] applied a ps pulsed laser to join fused silica glass and 304 stainless steel, obtaining a maximum bonding shear strength of 30.73MPa. These studies highlight the advancements in laser bonding using ultrashort pulsed lasers and their potential for achieving substantial shear strength in welds. However, despite these advantages, ensuring the necessary gap for laser welding between glass and metal is typically within a few micrometers, which can be challenging to achieve with industrial-grade techniques. Therefore, it is essential to carefully assess the appropriate welding method for aluminum-to-glass applications.

Using longer pulse lasers, such as milli- or nanosecond lasers, offers a more affordable alternative strategy while maintaining the benefits of LTB. For instance, Min et al. [16] investigated the influence of laser power, scanning speed, and pulse repetition frequency on the quality of the welds using a millisecond pulsed laser to join 6061 aluminum alloy and glass. They achieved a maximum bonding shear strength of 8.94MPa. Similarly, Floropoulos et al. [17] successfully performed LTB of aluminum to soda lime glass using a nanosecond (ns) pulsed laser, finding that the bond's maximum shear strength reached 12.01 MPa. In studies focused on aluminum alloy/non-metallic joints, Jiao et al. [18] examined how laser power and the number of laser scans affected bonding quality. They reported a maximum joining shear strength of 30.4MPa using an ns pulsed laser. These studies illustrate the feasibility of bonding aluminum with dissimilar materials using longer pulse lasers, provided that the appropriate laser processing parameters are carefully controlled. While these investigations demonstrate the potential of using cost-effective lasers for welding aluminum alloys and non-metallic materials, most research has primarily focused on individual parameters. This approach has hindered the ability to achieve optimal bonding strength, as it does not consider the combined effects of multiple parameters in the welding process. Consequently, identifying the optimal range of welding parameters based on the LTB mechanism to ensure good weld quality remains a challenge.

The creation of new compounds as a result of chemical bonds or reactions during the bonding process is often examined when assessing welding performance. For instance, Feng et al. [19] utilized SEM, EDS, and XPS analyses to specifically identify the formation of Cu-O-Si compounds and Cu₂O at the copper/glass interface during the ns pulsed laser welding of glass and copper. This highlighted their crucial role in enhancing weld strength. Similarly, Wang et al. [20] employed EDS and XPS analyses to demonstrate the development of a new TiSi system at the weld zone interface during millisecond pulsed laser of glass and titanium alloy. Additionally, a recent study by Zhang et al. [21] emphasized that the formation of a new amorphous zone, which combined aluminum and SiO₂ phases, significantly improved the quality of the joint while using an fs pulsed laser. However, it is important to note that most of these studies focused on analyzing the microstructure and nanostructure properties of the fracture surface of the weld, whether it involved glass or metal. Consequently, the exploration of the weld interfacial characteristic prior to separation to eliminate the effects of an external force is a challenge.

The laser power and repetition frequency are directly linked to single-pulse laser energy. This energy significantly affects the microstructure and bonding quality of the joints between glass and metal due to the energy absorbed in the area of the metal substrate that is irradiated by the laser [22]. Furthermore, each pulse reheats and remelts a portion of the previous spot that has not cooled sufficiently, meaning that the reheated area of the previous spot becomes the heat-affected zone (HAZ) for the next pulse. To characterize this aspect of pulsed laser welding, the overlapping factor is proposed, which considers the cumulative effect of multiple pulses hitting the same area [23]. Numerous studies have been carried out to demonstrate the impact of these parameters on weld quality. For instance, Tian et al. [24] used a ps pulsed laser to analyze the effects of single-pulse energy on microstructure and bonding strength of 304 stainless steel to glass, resulting in a maximum bonding strength of 21.7MPa. Zhan et al. [25] demonstrated that by reducing micro-defects in the fusion zone, the appropriate laser energy may enhance the bonding quality of aluminum to glass, achieving a maximum weld shear strength of 25.75MPa. Bu et al. [26] highlighted the impact of laser energy on the interfacial structure of 6061 Al alloy/non-metallic joints using an ns pulsed laser, resulting in an optimal bonding shear strength of 32.42MPa. Additionally, Wang et al. [27] showed that the pulse overlapping factor affects the bonding strength of glass to opaque materials, reaching a strength of 38.07MPa. A challenge in understanding the relationship between the laser joining process and bonding quality is that most studies have shown that the strengthening mechanisms primarily operate at the weld interfacial region. Additionally, the characteristics of interfaces with more complex shapes have not been thoroughly examined in relation to processing parameters. Consequently, the bond strength between metal and glass needs enhancement to satisfy the high-performance requirements in challenging operating environments.

The characterization of the weld interfacial zone, particularly in relation to the expansion of the mixing zone and the formation of the keyhole region, is critical for strengthening metal/glass joints [20]. Presently, only a limited number of studies indicate that the correlation between shape of the weld interfacial zone on the bonding quality of dissimilar materials. For instance, our previous investigations [28] effectively enhanced the strength of copper/glass joints by analyzing the shape of the bond interfacial region, which included shell-like, M-like, and hump-like formations in relation to laser energy. The strengthening mechanism involved the re-solidification of glass using ns pulsed technology. Additionally, Bu et al. [29] demonstrated that specific weld interface configurations, such as fishbone-like and nacre-like patterns, can significantly improve the bonding strength of 6061 aluminum alloy/non-metallic joints when using an ns pulsed laser. Consequently, to achieve high-quality heterogeneous joints, it is crucial to systematically study the growth of the weld zone with different shapes at the interfacial region during the laser joining of metal and glass.

Recent studies for glass/metal joints have mostly concentrated on assessing the manner in which processing parameters affect weld quality by examining results over a sizable welding area created by several overlapped line scans, which means that micro-defects are unavoidable in the weld seam. Obviously, this approach generates difficulties in determining the individual scanned line and its overlapping rate on glass/metal weld joints, which play a substantial role in sealing in MEMS technology or automotive manufacturing.

This study evaluates the mechanical characteristics and cross-sectional structure characterization of aluminum alloy/glass junctions using an ns pulsed laser corresponding to a single weld line. Our goal is to identify the ideal range of processing parameters suited to a ns pulsed laser's properties. By using SEM, high-resolution X-ray photoelectron spectroscopy (HR-XPS), and high-resolution transmission electron microscopy (HR-TEM) to characterize the microstructures, nanostructures, elemental composition, and phase configurations of the weld seam prior to separation, we are able to eliminate the effects of separation force and obtain insights into how welding parameters affect weld morphologies within and around the weld seam's periphery. Additionally, the shear strength of the welded joints is characterized by the expansion keyhole zone in correlation with the laser overlapping factor, single-pulse energy, weld seam interface geometries, and fracture surface morphology. This analysis aims to enhance the development and application of nanosecond laser technology for joining transparent materials and metals.

2. Materials and experimental procedure

2.1. Material preparation

The 5052 aluminum alloy (5052AA) sheet and commercially available EAGLE XG® glass were utilized in the investigation. [Table 1](#) lists each of their chemical compositions. The glass sample used in the experiments had a thickness of 0.7 mm and was cut into a rectangular form that measured 75 mm in length and 25 mm in width. The experimental aluminum workpieces were made to match the glass in terms of length and width, but they were 3 mm thick. Numerous material characteristics, including melting temperature, density, specific heat capacity, thermal expansion, and thermal conductivity, have a substantial impact on the thermal process in laser welding. The mechanical and thermal characteristics of the 5052AA plates and EAGLE XG® glass used in this investigation are displayed in [Table 2](#).

Table 1. Chemical composition of the EAGLE XG® glass and 5052AA aluminum alloy.

Materials	Element (wt.%)						
	SiO ₂	Al ₂ O ₃	B ₂ O ₃	MgO	CaO	SrO	BaO
EAGLE XG® glass	59.7	17.11	12.27	1.39	7.33	0.81	0.01
5052AA aluminum alloy	Al	Mg	Cr	Cu	Fe	Si	Zn
	97.2	2.5	≤0.25	≤0.1	≤0.4	≤0.25	≤0.1

Table 2. Thermal and mechanical properties of 5052AA aluminum alloy and EAGLE XG® glass.

Material properties	5052AA aluminum alloy	EAGLE XG® glass
Coefficient of thermal expansion (1/K)	23.8×10 ⁻⁶	3.2×10 ⁻⁶
Thermal conductivity ((W/(m.K))	138 (25°C)	1.09 (23°C)
Melting point (°C)	607	669 (Strain Point)
		772 (Annealing point)
		971 (Softening Point)

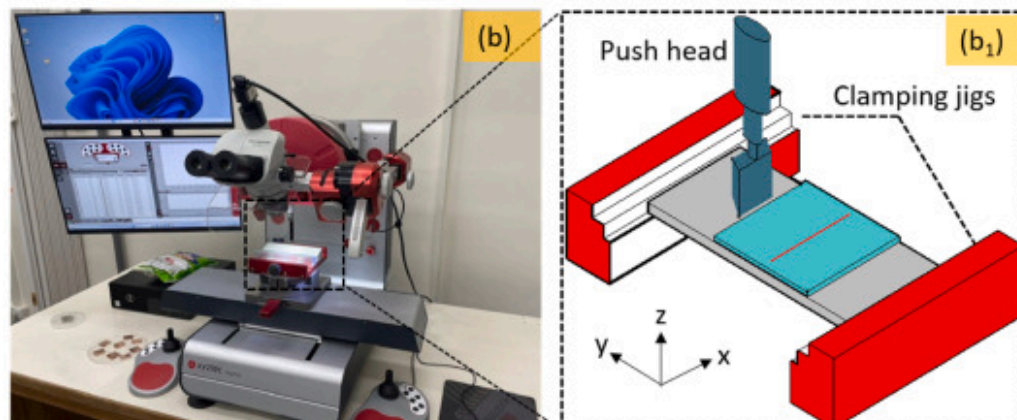
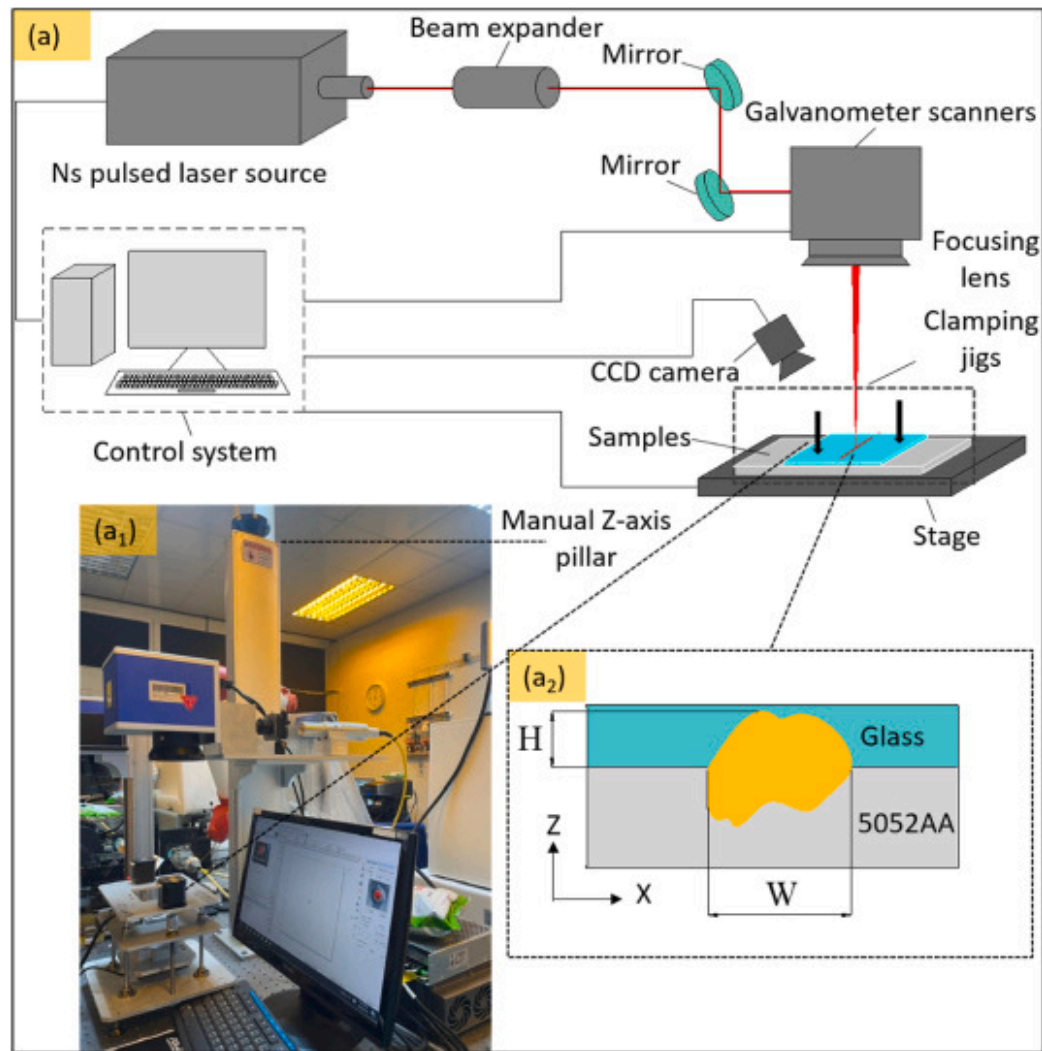
Material properties	5052AA aluminum alloy	EAGLE XG® glass
		1293 (Working point)
Density (g/cm ³)	2.75	2.38

The initial surface roughness of the as-purchased aluminum alloy plates varies from one another, approximately with $R_a=2.5\mu\text{m}$. The surface roughness of the EAGLE XG® glass substrate, as purchased, was approximately $R_a=0.15\mu\text{m}$. After thoroughly cleaning the surfaces of the glass and aluminum alloy with isopropyl alcohol ($\text{C}_3\text{H}_8\text{O}$), they were ultrasonically agitated for 25 min and then carefully rinsed with acetone. Following a deionised water rinse, nitrogen gas was used to carefully dry the surfaces.

2.2. Laser welding setup

Maintaining a very small gap between the two surfaces that will be welded is crucial for producing high-quality welds. In our project [28], we constructed a clamp structure that would allow precise control of the applied force using two stepper motors (Model 42BYGH47) and a load cell (Model HX711). One of the three ball-screw frames of the device was powered by stepper motors to hold the other two in place. The second frame was movable forward or downward using two screws. In this investigation, two surfaces were assembled using a clamping force of 68N prior to the welding procedure.

As illustrated in Fig. 1(a), a pulsed fiber laser (YLP-1-100-20-20, IPG) that emits near-infrared light with a wavelength of 1070nm and a pulse duration of 100 ns was employed for all laser welding investigations. Its greatest average power is 20W. Additionally, the welding system includes a mirror for laser beam supply, a focusing lens, and an X-Y workable. The focal distance of the focusing lens was 290mm, and the laser beam was focused on a $50\mu\text{m}$ -diameter point on the focal plane. Remarkably, vertical focus plane adjustment was possible with the Galvanometer scanner connected to a manual Z-Axis 800mm pillar. This allowed for exact control of the laser spot's placement along the Z-axis. A coaxial CCD camera was also used to track the laser point's position on the focus plane. Fig. 1 (a₁) clearly illustrates the nanosecond laser machining system, featuring a clamping sample system and a manual Z-axis pillar for controlling the focal plane position.



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Fig. 1. Laser transmission bonding and weld shear strength measurement experimental setup: Fig. 1 (a) shows a schematic illustration of the ns pulsed laser processing system used in this study. Fig. (a₁) Image of a nanosecond laser machining system with clamping sample system and a manual Z-axis pillar to control the focal plane position; Fig. (a₂) Cross-sectional view of the weld line, emphasizing the high (H) and width (W) geometric terms along with their referential coordinates. Fig. 1 (b) Photograph of a tensile tester used to measure shear-tensile separation force. Insert: (b₁) A schematic illustration of the jigs used for clamping weld joints during shear force testing.

2.3. Characterizations

2.3.1. Weld line properties

To ensure a thorough and systematic comparison of the weld properties produced using various laser processing parameters, we conducted the welding procedure as a line weld with a precise length, L , of 18mm. To effectively eliminate potential end effects associated with the starting and ending points of the laser irradiation during the welding process, we concentrated our analysis on the central portion of the weld [22]. This approach guarantees that the evaluated properties are representative of the main welding region, thereby minimizing any variations or distortions that could arise from the initiation and termination points, and ultimately enhancing the reliability of our findings.

Characterizing the weld area and surface topography requires an examination of the cross-sectional surface, which is perpendicular to the weld line. Specifically, the X–Z plane shown in Fig. 1 (a₂) provides a view for analysis, allowing us to assess the extent of the weld and gain insights into various aspects of the surface features and morphology of the welded joint. Analyzing the cross-sectional image reveals valuable information regarding the structural integrity, joining quality, and overall appearance of the weld. For the transverse cut of the aluminum alloy/glass weld seam, we used an IsoMet™ 4000 Linear Precision Saw in combination with EXTEC diamond wafering blades. To ensure optimal cutting conditions and eliminate damage

or distortion to the samples, we utilized distilled water mixed with Buehler Cool 3 Cutting Fluid (33.8 oz) for cooling during the cutting process.

To improve image quality, we utilized grinding and polishing techniques with the Buehler Ecomet 30 Semi-Automatic Grinder Polisher. The samples were ground sequentially using abrasive grinding papers with grit sizes of P1500, P2000, and P4000 at a rotational speed of 100rpm. After grinding, the surfaces were polished at an average speed of 40rpm. Once polishing was complete, the samples were thoroughly cleaned with ethyl alcohol (C_2H_5OH) and subjected to a 15-min ultrasonic wash to remove any dirt or contaminants. Finally, cross-sectional surface images were captured using the Confocal Laser Scanning Microscopy (CLSM) technique with the Keyence VK-X1000.

Using high-resolution images obtained from CLSM, we carefully assessed the dimensions of the weld seam by referencing [Fig. 1](#) (a_2) to measure the width (W) and height (H). The original optical images of the welding seam were examined at a magnification of 50x. We deliberately selected and marked specific positions within the interface area that were pertinent to the critical weld dimensions W and H . To ensure a thorough and accurate assessment of the weld seam's geometry, we analyzed five cross-sectional areas to calculate the mean values for both seam width and height. After measuring the width (W), we calculated the projected surface area (S), using formula $S \approx L \times W$, which represents the 2D area of the weld seam. Meanwhile, the measurement outcomes of the height (H) will be discussed in [Section 3.2.2](#). This meticulous approach not only guarantees precision but also provides essential dimensional insights for further analysis, highlighting the importance of our findings.

2.3.2. Welding shear strength

In this experiment, the bonding strength of the weld was assessed using the fracture point of its highest sustainable separation shear force, as measured by an XYZTEC Condor Sigma Series: Condor Sigma. This gadget has the ability to raise existing standards because of its six sensors arrangement and maximum tensile-shear force of 1000N. The measurement device used to determine the maximum tensile-shear force is seen in [Fig. 1\(b\)](#).

2.3.3. Weld morphology

In order to examine the surface morphology of the weld seam in the cross-sectional plane, the welded samples were cut with EXTEC Diamond Wafering Blades and an IsoMet™ 4000 Linear Precision Saw. A Buehler Ecomet 30 Semi-Automatic Grinder Polisher was then used to polish the cut samples with $0.3\mu m$ Al_2O_3 powder in order to evaluate the cross-sectional region's

surface topography. The polished samples underwent a rigorous cleaning procedure using isopropyl alcohol (C_3H_8O) and a 15-min ultrasonic wash to remove any dirt or contaminants. Following the cleaning process, cross-sectional surface images were acquired with an SEM (JOEL, JSM-7000F) and a CLSM (Keyence VK-X1000).

The elemental compositions in the 5052AA/ SiO_2 weld zone were examined using HR-XPS in addition to surface images. In order to analyze nanoscale morphologies and structures at the interfacial level, samples were sectioned around the welded aluminum alloy/glass interface using the dual-beam focused ion beam (FIB, Versa 3D) technique. High-resolution transmission electron microscopy (HR-TEM, JEOL, JEM-2100) was used to analyze the weld zone in order to find any potentially newly produced phases.

3. Results and discussion

3.1. Essential factors in LTB for aluminum alloy/glass joints with an ns pulsed laser

The final welding quality and the shape of welded joints between glass and aluminum alloy are greatly impacted by a number of laser processing parameters used during the LTB, in addition to the initial roughness of the glass and aluminum surfaces. These critical parameters include average laser power, P , pulse repetition frequency, f , scanning speed, v , focal plane position, pulse overlapping factor, PO , single-pulse energy, E_s (Table 3).

Table 3. Laser processing parameters.

Experiment level	1	2	3	4
Average power (W)	4.78	6.72	8.79	10.81
Scanning speed (mm/s)	5	50	100	150
Frequency (kHz)	20			

The combined dynamic influence of these elements is significant in the actual welding process, where light interacts with the material. Instead of achieving mathematical independence, this leads to their interconnectedness. When the non-linear effects of the material during processing are taken into account, such as nonlinear light-material interactions, temperature-dependent

changes in properties, phase-transition-induced structural and property changes, and other complex factors, the phenomena involved become noticeably more non-linear. The brief duration of ns pulsed laser pulses exacerbates this complexity even more. Understanding and regulating the welding process is difficult due to the complexities created by the interaction of various components, necessitating a thorough strategy to achieve the best results.

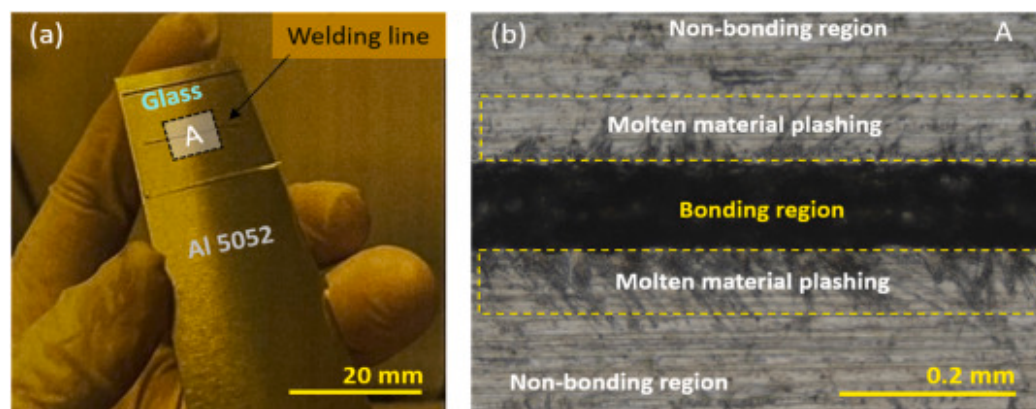
We kept a few constants throughout this investigation to simplify the complexities and define an appropriate range of welding parameters (processing window) for our existing system. Utilising the interface position, also known as the zero position, offers the optimal energy distribution balance and enables efficient mixing of the two materials at the interfacial area. Specifically, during laser irradiation, a constant jig clamping force (CF) of 68N was guaranteed for optical contact conditions between glass and aluminum alloy [28]. P and v were the variables that were changed.

Numerous test findings in this study demonstrated the relatively small parameter window that can be used for welding within the existing laser system's capabilities. While average laser power levels above 12.81 W led to poor homogeneity and problems with structural integrity, those below 6.72 W proved insufficient for efficient welding. Thus, four average laser power levels, $P=6.72, 8.79, 10.81$, and 12.81 W-were used for this investigation. The laser energy becomes excessive when the repetition frequency is less than 20kHz, leading to severe thermal stress and unsatisfactory bonding. The interfacial region cannot be melted by the laser energy when the repetition rate is higher than 20kHz. At the same time, we investigated the interaction between $f=20$ kHz and four levels of scanning speed: $v=5, 50, 100$, and 150 mm/s, respectively. This combination produced a pulse overlapping factor $PO = \left(1 - \frac{v}{D \times f}\right) \times 100$ (%) of the beam spot, which may melt and mix the two interfacial materials together at a minimum value of 85%. Increased overlapping could provide more chances to promote chemical reactions and inter-diffusion, controlling microstructures within the weld and surrounding the glass/aluminum alloy interface. The pulse overlapping factors for the four scanning speeds were 85, 90, 95, and 99%, respectively. In addition, the single-pulse energy, denoted as E_s , $E_s = \frac{P}{f}$ is a commonly used metric in welding that incorporates incident laser power and laser frequency. In this investigation, the single-pulse energy measured 0.336, 0.439, 0.541, and 0.641 mJ.

It is crucial to optimise the focus plane position. Placing it directly on the glass interface causes substantially less effective penetration due to transmission. Conversely, energy dispersion due to the focal point beneath the contact may result in weaker bonding. A defocus distance of zero position from the interface was determined to be suitable for our study's laser system after several experimental tests. There are two possible explanations for this: First, by placing the focus point at the interface, both substrates can get direct energy application, guaranteeing that the bond region gets the ideal amount of laser energy. Second, the

zero position may make it easier for molten aluminum to enter the surface imperfections of roughened aluminum substrates, improving adherence.

Before the joints' shear strength is tested, Fig. 2 shows an optical microscope (OM) image of a well-formed weld seam between the glass substrate and the aluminum alloy plate. The joining procedure was conducted under the parameters $P=8.79\text{W}$, $v=50\text{mm/s}$, $E_s=0.439\text{mJ}$, $PO=90\%$. Interestingly, a close inspection reveals no obvious defects that could be seen with the naked eye, as seen in section A's inset. This demonstrates how well-joined a weld seam can be produced using the present laser processing equipment. Unlike ps- or fs pulsed lasers used for welding glass and metal, glass can directly absorb laser energy through a non-linear absorption mechanism [25]. Nd:YAG laser light is highly transparent to glass, with over 90% of the light being transmitted. Meanwhile, in the induction welding of dissimilar materials, the formation of the molten zone is achieved through a combination of eddy current induction, the compressive force exerted by a consolidation roller, and heat transferred to the surrounding air from forced convection caused by air cooling nozzles, ultimately resulting in mechanical bonding [9].



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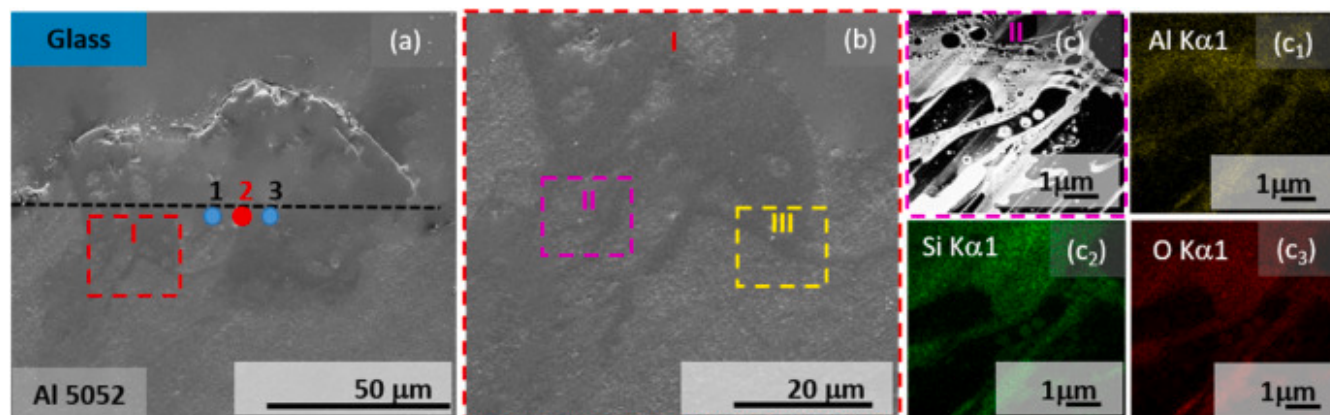
Fig. 2. Images from the CLSM's optical microscope (OM) demonstrate that line-bonded seams can be successfully achieved using the following processing parameters: $P=8.79\text{W}$, $v=50\text{mm/s}$, $f=20\text{kHz}$, $PO=85\%$, and $E_s=0.439\text{mJ}$. Fig. 2 (a) shows an overview of a line welding seam measuring 18mm. Fig. 2 (b) indicates an enlarged picture demonstrating the superior surface morphology attained.

Conversely, in ns pulsed laser welding of glass and aluminum alloy, the primary joining mechanism involves the laser penetrating the glass and being absorbed by the aluminum alloy surface. This absorption raises the temperature of the aluminum, leading to phase changes at the glass-aluminum alloy interface. The result is local melting followed by solidification, which forms a strong joint as in Fig. 2 (b). Furthermore, weld pool dynamics are affected by the surface roughness of the aluminum alloy. Molten material splashing along the welding line is more likely when irregularities cause unstable melt pools.

3.2. Cross-sectional structures analysis

3.2.1. Microstructure characteristics

SEM pictures in Fig. 3 (a), (b), and (c) show cross-sectional views of the aluminum alloy/glass weld seam. The processing parameters of $P=8.79\text{W}$, $v=50\text{mm/s}$, $f=20\text{kHz}$, $PO=85\%$, $E_s=0.439\text{mJ}$. Fig. 3 (a) illustrates the absence of substantial defects, such as cavities or micro-pores, within the weld zone, which is consistent with the overall weld structure represented in Fig. 2 (a). This cross-sectional image emphasises the primary composition of the weld zone, which was glass. This observation is influenced by two main reasons. Initially, the local 5052AA temperature rose dramatically as the aluminum alloy surrounding the glass/aluminum alloy interface absorbed laser irradiation energy. This resulted in energy transfer to the glass and another area of the aluminum alloy through the heat conduction phenomenon. This procedure may cause the glass to soften and then melt, causing the aluminum alloy and glass to melt together. Second, enlarging the glass's molten zone was facilitated by the incident laser coming from the glass side and the nonlinear optical absorptivity of high-temperature softened glass. The opacity of the glass to incident laser light increased with temperature, leading to both higher absorption by the glass and additional enlargement of the molten zone within the glass [22]. Specifically, the absorption along the incoming beam passage resulted in the formation of many hot spots on the glass side. Their surroundings experienced increased temperatures and pressures as a result of this production, which helped to build the M-sharped re-solidified glass region seen in Fig. 3 (a) [28].



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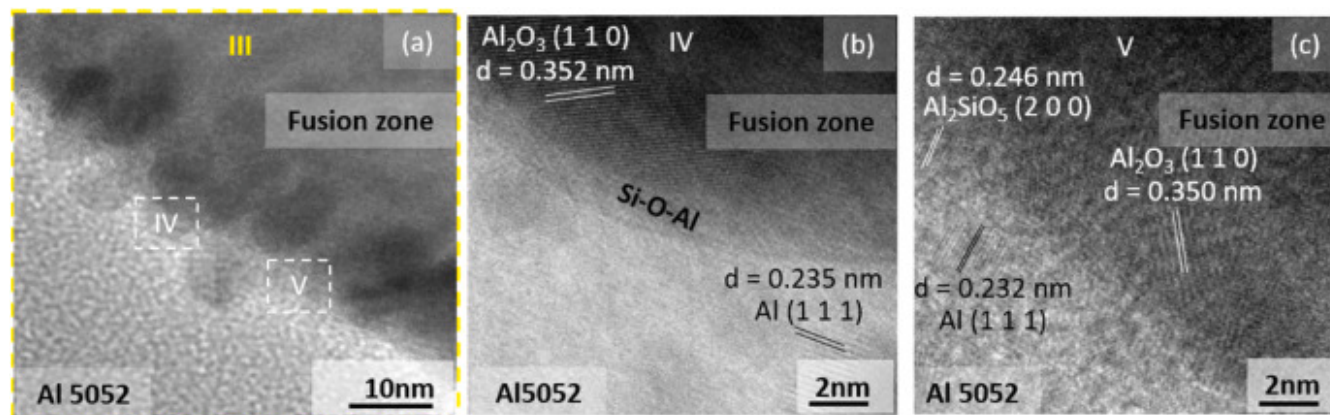
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Fig. 3. Typical cross-sectional SEM surface morphologies that show interior microstructures in the weld seam were created with the following processing parameter sets: $P=8.79\text{W}$, $v=50\text{mm/s}$, $f=20\text{kHz}$, $PO=85\%$, and $E_s=0.439\text{mJ}$. Fig. 3 (a) observes an SEM image revealing the mixing of aluminum and glass. Fig. 3 (b) indicates a higher-resolution SEM image for the enclosed (I) area in Fig. 3 (a). Fig. 3 (c) shows an SEM magnified image for the pink-dashed box labelled (II) region in Fig. 3 (b). Fig. 3(c₁), (c₂), and (c₃) represent the EDS mappings of Al, Si, and O elements, respectively corresponding to Fig. 3 (c). The positions from (1), (2), (3) in Fig. 3 (a) denote particular locations for carrying out XPS analysis; the yellow-dashed box labelled (III) in Fig. 3 (b) denote a specific zone for nanostructure observation in Fig. 4. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

A keyhole zone is produced in Fig. 3(b) when the molten zone develops towards the aluminum alloy side. The following is the formation mechanism: molten aluminum can evaporate when the temperature rises above the aluminum alloy's vaporization point due to an energy density that is high enough. A high-pressure vapour plume created by this vaporization pushes the molten pool outward, causing a cavity to form in the aluminum alloy. Furthermore, the keyhole's capacity to absorb, scatter, and refract the high-intensity laser beam can create a plasma plume, which has a substantial impact on the laser's energy distribution. As seen in Fig. 3(b), these interactions change the course and intensity of the beam, creating a modified keyhole zone on the 5052AA side that resembles a root. Following the development of the modified keyhole zone, which has a root-like shape, the incident laser can be reflected several times within the weld zone via the reverse Bremsstrahlung mechanism and

Fresnel reflections, greatly increasing energy absorption. The reverse Bremsstrahlung mechanism occurs when high-energy electrons, generated by laser-material interaction, decelerate near atomic nuclei, emitting radiation. This radiation propagates opposite the incident laser, dissipating energy while enhancing heat deposition in the keyhole. Meanwhile, Fresnel reflection describes the laser's reflection at material surfaces. In the keyhole, multiple reflections increase energy absorption in the weld region [20]. Additionally, the expansion of the keyhole area produces a mechanical interlocking effect that raises the glass/aluminum alloy's breaking shear strength.

Fig. 3 (c) clearly shows the contrast between the dark and light regions, which correspond to single-crystal Al, Si, or O nanoparticles as well as a mixture of Al and glass components. Several factors contribute to the area's difference: Because Al has a greater atomic number than the primary components of glass, it interacts with the electron beam to produce more secondary electrons, giving the impression of brightness. But the glass family's lighter components, such as oxygen, lessen secondary electron emission overall, making the glass side appear darker. Conversely, the melting and mixing of many components result in the formation of the molten zone. HR-XPS investigation, as covered in Section 3.2.3, confirms that this region may contain intermetallic compounds (Al-Si), Al-O, or Si-O-Al layers that result in non-uniform dark and light areas at the molten region, as shown in Fig. 3 (c). These compounds' presence in the interfacial region is essential for strengthening the connection between 5052AA and SiO_2 . A more thorough examination of the nanostructure in this area will be conducted later using HR-TEM observations, as shown in Fig. 4 (a), (b), and (c).



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Fig. 4. Interfacial TEM images illustrating internal nanostructures within the weld seam, produced using the processing parameter sets of $P=8.79\text{W}$, $v=50\text{mm/s}$, $f=20\text{kHz}$, $PO=85\%$, $E_s=0.439\text{mJ}$. Fig. 4 (a) shows an HR-TEM image enclosed by a yellow-dashed box labelled (III) in Fig. 3 (b). Fig. 4 (b) and (c) observe the HR-TEM images of a zone (IV) and (V), respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Due to variations in composition gradients and local pressure variations, convectional intra-mixing was the predominant mechanism for mixing within the molten zone throughout this phase [30]. The Marangoni effect [31] and thermal buoyancy are two more commonly mentioned theories for improving mixing in the weld pool. Nevertheless, these mechanisms had comparatively less of an impact in the setting of this investigation. The main explanation is that temperature variations at the interface between distinct phases are what essentially causes the Marangoni effect. This phenomenon might theoretically occur in situations where there is low mutual solubility, such as at the interface between air and an uncovered melt pool. However, the interface impact was less significant since the molten aluminum alloy could blend and diffuse with the molten glass. Furthermore, creating sizable convection currents for mixing in this work was a problem for thermal buoyancy because of the small size of the molten pool (at the microscale) and the high viscosity of the liquid metal and molten glass.

The EDS mappings for the Al, Si, and O elements, respectively, in the characterised region shown in Fig. 3 (c), are shown in Fig. 3 (c_1), (c_2), and (c_3). This section matched the boxed region's magnified view in Fig. 3 (b) inset. The maps showed that blocky mixed areas of glass and aluminum alloy made up the majority of the structure surrounding the interface. This demonstrated that the primary mechanism for mixing within the molten pool during the welding operation was convectional intra-mixing between the aluminum alloy and glass. The primary causes of this convection were identified as non-uniformly localised high pressure, which was brought on by the quick rise in temperature brought on by the incident laser energy that was absorbed. Convectional mixing is further aided by spatial differences in temperature and concentration. Furthermore, Fig. 3 (c_2) and (c_3) demonstrate that the distributions of Si and O elements exhibited a high degree of spatial consistency. This observation suggested that, in the majority of the glass present in the weld region, the atomic structure remained intact and was not disrupted during the welding process.

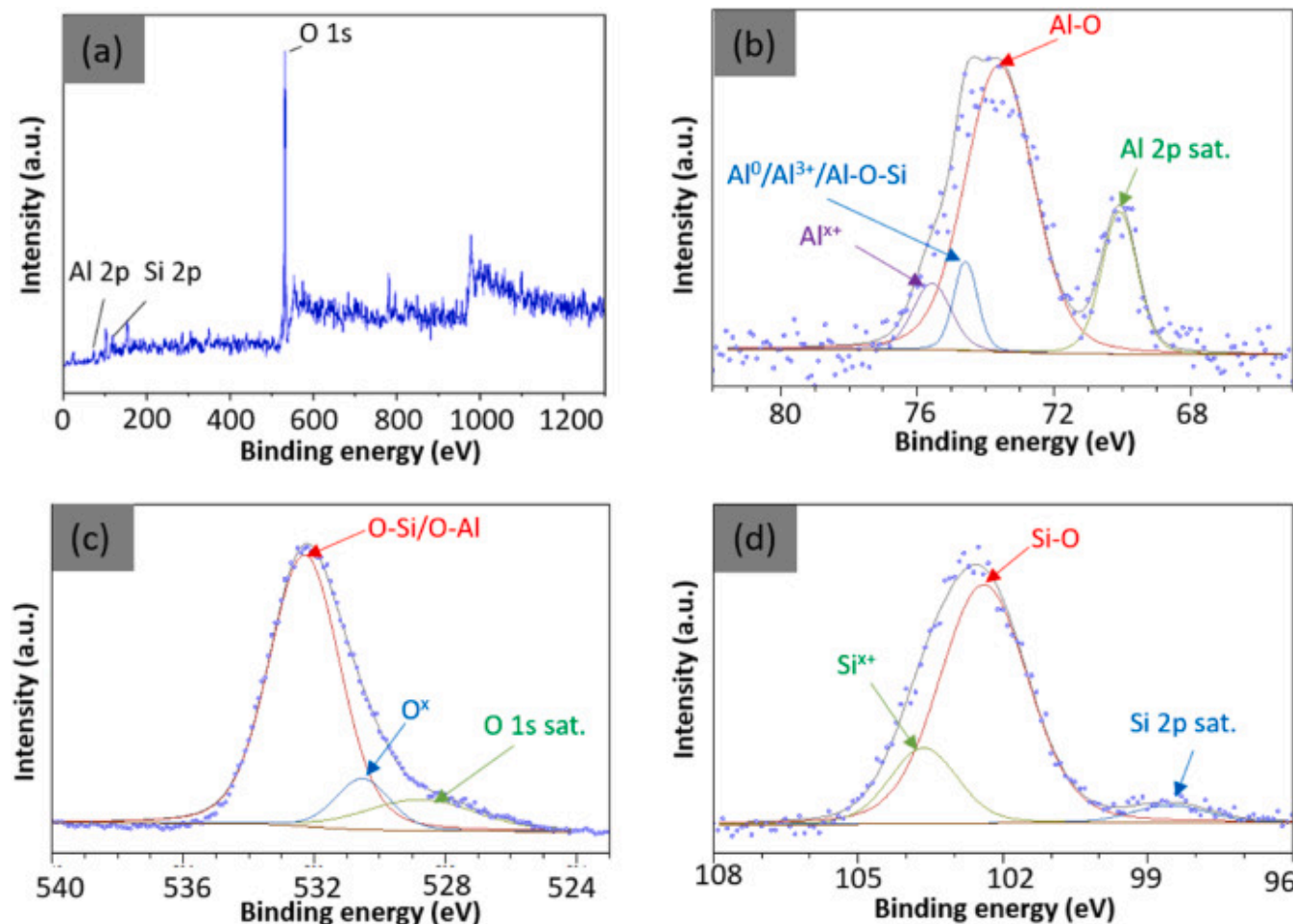
3.2.2. Nanostructure observation

Region III was chosen for additional weld nanostructure examination. Fig. 4 (a) shows the HR-TEM picture of Region III, where Fig. 4 (b) and (c) analyze two different interfacial positions, IV and V, respectively. The greyscale intensity in this observation allowed for the distinction of lighter and darker areas between the aluminum alloy and mixing zone sides. Moving on to Fig. 4

(b), it is easy to identify a single crystalline structure, Al (111), with a d-spacing of 0.235 nm that corresponds to the face-centred cubic structure of Al (JCPDS-ICDD no. 00004–0787, $d_{111}=0.233$ nm) and the structure of aluminum oxide, Al_2O_3 (110), which has a d-spacing of 0.352 nm. This number is in good agreement with the face-centred cubic structure of Al_2O_3 (JCPDS-ICDD no. 76–0144, $d_{110}=0.3480$ nm). Additionally, Region IV, which has a hazy arc form between the two areas, is where the Si-O-Al region is observed. As a result, throughout the welding process, dynamic reactions at the 5052AA/ SiO_2 interface were promoted by the incident laser energy. The noticeable change in the occurrence of the new compound Si-O-Al, which is present at the interface of amorphous glass and aluminum nanoparticles over most of the weld's breadth, offers strong proof of a strong chemical reaction among the weld's ingredients. However, distinct areas of the 5052AA/ SiO_2 interface show variations in the size of the newly produced compounds. At the same time, Fig. 4 (c) shows that Al_2O_3 (110) and single nano-crystalline Al (111) are continually present, with d-spacings of 0.232 nm and 0.350 nm, respectively. When compared to Fig. 4 (b), the emergence of alumino silicate polymorphs, Al_2SiO_5 , along the boundary, which corresponds to a d-spacing of 0.246 nm, showed that a redox reaction may occur between 5052AA and SiO_2 at the interfacial zone. Furthermore, in accordance with the face-centred cubic structure of Al_2SiO_5 (JCPDS-ICDD no. 22–0018, $d_{200}=0.238$ nm).

3.2.3. Elemental analysis

Strong chemical bonding between the elements in the weld zone is a critical prerequisite for improving welding performance. The elemental composition, chemical state, and electronic state resulting from chemical interactions between the elements Al, O, and Si were all examined in this study using an XPS test. As shown in Fig. 3, X-ray measurements were performed with a spot size of 400 μm , and readings were averaged from five different centre places on the cross-sectional plane of the weld zone, perpendicular to the line weld seam, in accordance with the methods described in Ref. [28]. In this study, the XPS spectra were obtained close to position No. 3, as shown in Fig. 3 (a), which is consistent with the findings in Fig. 5(a), (b), (c), and (d). Fig. 5 (a) shows the survey scan of the samples that were taken with the Al K α beam. Peaks representing the spectrum lines of Al 2p, O 1s, and Si 2p were seen at the interface between the aluminum alloy and glass. These peaks had binding energies of roughly 76, 530, and 101 eV, respectively.



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Fig. 5. High-resolution XPS spectra, including the full survey spectrum (a), Al 2p spectrum (b), O 1s spectrum (c), and Si 2p spectrum (d). These spectra correspond to the processing parameter set: $P=8.79\text{W}$, $v=50\text{mm/s}$, $f=20\text{kHz}$, $PO=85\%$, $E_s=0.439\text{mJ}$. The purple dots represent measured data points, while the purple lines depict the fitting lines generated using CasaXPS software. The corresponding compositions identified from Al 2p, O 1s, and Si 2p are indicated by arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

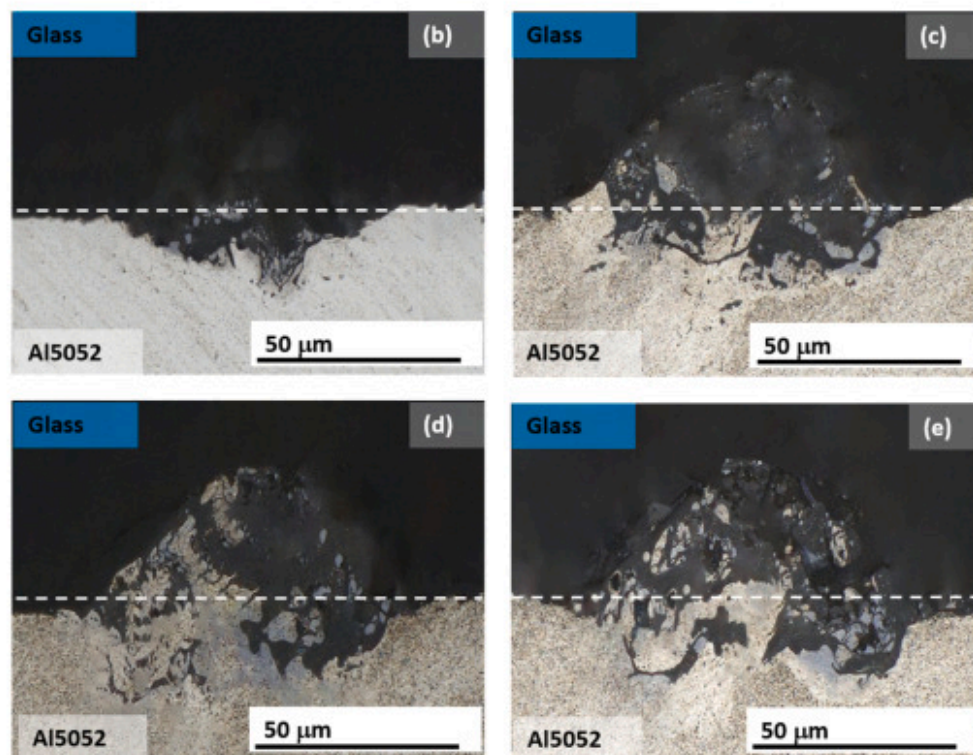
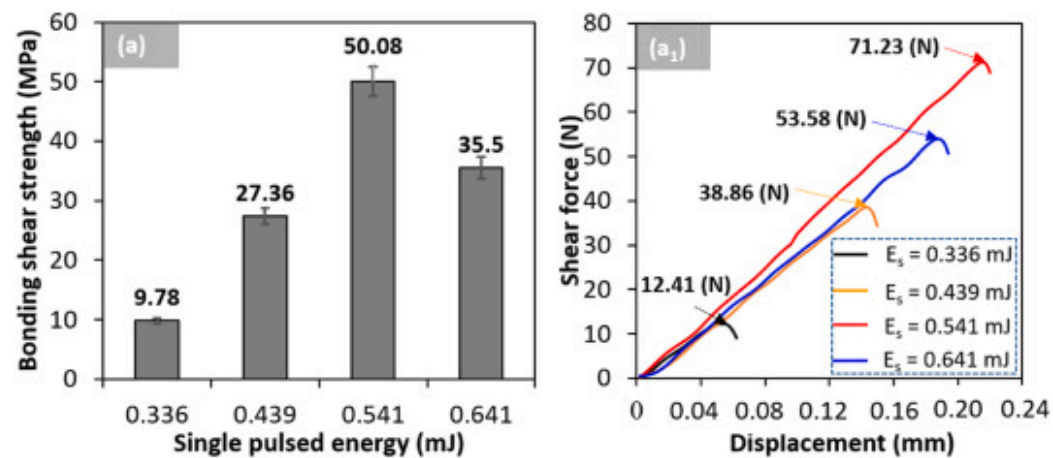
According to Fig. 5 (b), the peak at 73.24eV was ascribed to Al 2p, with the Al-O combination potentially serving as a contributing source phase [32]. Interestingly, the Al^{x+} state, which has energies 2.46eV higher than the Al 2p peaks, is responsible for the band at 75.7eV [33]. Shake-up satellite peaks at the bonding energy were found to be around 4.3eV lower than the $\text{Al}^0/\text{Al-O}/\text{Si-O-Al}$ phases [34,35].

Continuing to examine the unique elemental properties of O 1s, which has a binding energy of 531.5eV (Fig. 5(c)), and taking into account the significant amount of oxygen in the glass family, the O-Al or O-Si combinations were identified as the source of the noticeable peak at 532.9eV [35,36]. The Si-O-Al mixed oxide state in the sample may occur concurrently, leading to the O^x state, which is particularly represented by the peak centred at 531.7eV. Shake-up satellite peaks at the bonding energy were found to be around 4.6eV lower than the O-Al/O-Si peaks. The high-resolution line of Si 2p in Fig. 5(d) shows a double extraction peak at 102.2eV and 103.3eV, which represent Si-O and Si^{x+} , respectively [37]. Furthermore, at the welding energy, the shake-up satellite peaks were seen at a distance of roughly 4.8eV less than the Si^{x+} peak.

3.3. Influence of processing parameters on the weld characteristics

The strength of the welded aluminum alloy/glass junctions was assessed using the shear force separation test. The weld strength was measured using a formula $\sigma = \frac{F_{\max}}{S}$, where $F_{\max}$ represents the maximum shear force leading to the separation failure of the bond, and S is the projected surface area of the welded seam [22,28].

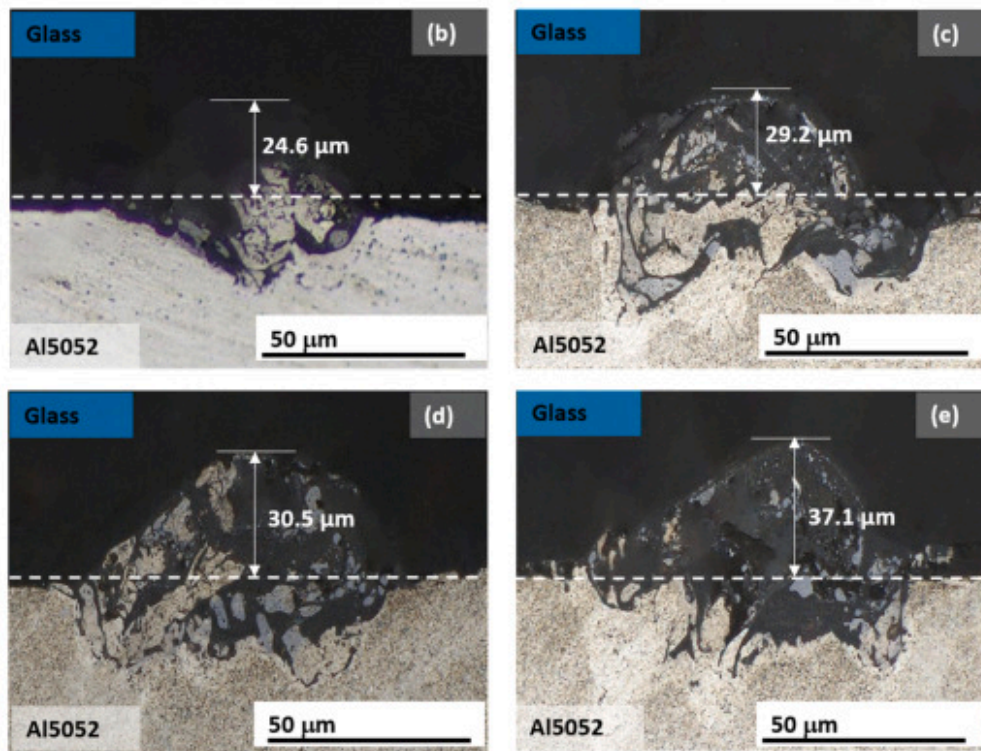
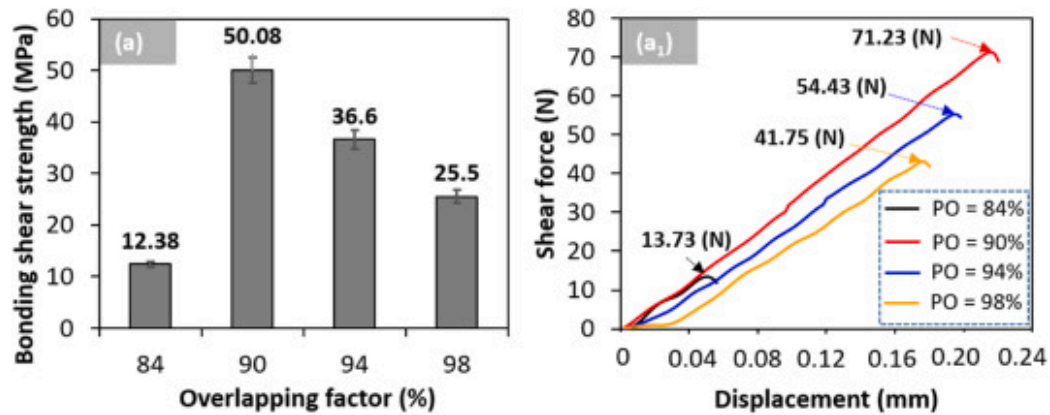
The weld seam's cross-sectional surface morphology, as seen in Fig. 6, Fig. 7, is re-examined in order to evaluate the impact of two specific processing factors on the weld zone's shape and associated properties: single-pulse energy and laser pulse overlapping factor.



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Fig. 6. The relationship between the microstructure properties of 5052AA/glass joints, bonding strength, and single-pulse energy. Fig. 6 (a) provides a comprehensive analysis of bonding shear strength as a function of single-pulse energy, based on precise tensile shear-force separation testing. The force-displacement curves illustrated in Fig. 6 (a₁) clearly demonstrate the performance of bonding joints at different single-pulse energy levels. Namely, the black, orange, red, and blue lines correspond to single-pulse energy values of 0.336 mJ ($P=6.72\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$), 0.439 mJ ($P=8.79\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$), 0.541 mJ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$), and 0.641 mJ ($P=12.81\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$), respectively. The maximum shear force values according to various single-pulse energies are indicated by arrows. Furthermore, Fig. 6 (b), (c), (d), and (e) offer detailed CLSM images that capture the cross-sectional surfaces of the weld seam at each single-pulse energy level, highlighting the unique effects of varying energy settings. The corresponding processing parameters are meticulously set: (b) $E_s=0.336\text{ mJ}$ ($P=6.72\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$); (c) $E_s=0.439\text{ mJ}$ ($P=8.79\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$); (d) $E_s=0.541\text{ mJ}$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$); (e) $E_s=0.641\text{ mJ}$ ($P=12.81\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $PO=90\%$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



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Fig. 7. Correlation between bonding strength, microstructural properties of 5052 aluminum alloy/glass junctions, and the pulse overlapping factor. Fig. 7 (a) provides a comprehensive analysis of bonding shear strength as a function of pulse overlapping

factor, based on precise tensile shear-force separation testing. The force-displacement curves showcased in Fig. 7 (a₁) reflect the impressive performance of bonding joints across various pulse overlapping factor values. Specifically, the black, orange, red, and blue lines illustrate pulse overlapping factor values of $PO=84\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=5\text{ mm/s}$, $E_s=0.541\text{ mJ}$), $PO=90\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $E_s=0.541\text{ mJ}$), $PO=94\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=100\text{ mm/s}$, $E_s=0.541\text{ mJ}$), $PO=98\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=150\text{ mm/s}$, $E_s=0.541\text{ mJ}$). The maximum shear force values according to various pulse overlapping factors are indicated by arrows. Additionally, Fig. 7 (b), (c), (d), and (e) present captivating CLSM images that highlight the cross-sectional surfaces of the weld seam at various pulse overlapping factors. The parameters for processing have been precisely calibrated: (b) $PO=84\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=5\text{ mm/s}$, $E_s=0.541\text{ mJ}$); (c) $PO=90\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=50\text{ mm/s}$, $E_s=0.541\text{ mJ}$); (d) $PO=94\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=100\text{ mm/s}$, $E_s=0.541\text{ mJ}$); (e) $PO=98\%$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $v=150\text{ mm/s}$, $E_s=0.541\text{ mJ}$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.3.1. Influence of single-pulse energy

Fig. 6 (a) illustrates the impact of single-pulse energy on the shear strength of aluminum alloy/glass joints. Processing parameters included a fixed pulse overlapping factor (PO) of 90% and a frequency of $f=20\text{ kHz}$. Meanwhile, $P=6.72, 8.79, 10.81$, and 12.81 W were the selected power levels. Fig. 6 (a) shows the resulting shear strengths for these examples, which show values of $\sigma=9.78, 27.36, 50.08$, and 35.50 MPa , respectively. Notably, the initial bond strength was comparatively low at 9.78 MPa at $E_s=0.336\text{ mJ}$, but after 1.3 times the addition of single-pulse energy, it significantly increased to 27.36 MPa . The bond strength improved even more, reaching 50.08 MPa , while E_s rose by 0.541 mJ . Nevertheless, $E_s=0.641\text{ mJ}$ σ decreased to 35.5 MPa .

Fig. 6 (a₁) presents the force-displacement curves for different single-pulse energy levels. The shear force of the 5052 aluminum alloy/glass joints showed a significant decline after reaching maximum values of 12.41 N , 38.86 N , 71.23 N , and 53.58 N . These values correspond to single-pulse energies of 0.336 mJ , 0.439 mJ , 0.541 mJ , and 0.641 mJ , respectively. The maximum displacements of the welding zone for the various energy levels were recorded at 0.057 mm , 0.156 mm , 0.219 mm , and 0.189 mm . To validate the bonding shear strength values shown in Fig. 6 (a), the mean projected surface area of the welded seam (S) was measured at 1.269 mm^2 , 1.420 mm^2 , 1.422 mm^2 , and 1.509 mm^2 , corresponding to the different single-pulse energy levels. Notably, the maximum shear force of the weld zone increases when the single-pulse energy rises from 0.336 mJ to 0.541 mJ . The shape of the welding seam interface varied with increased laser energy, which likely contributed to the rise in the maximum force of the bonded joint [28]. When E_s increases to 0.641 mJ , the maximum shear force significantly decreases. This reduction can be attributed to the increasing laser absorption coefficient of glass at higher temperatures, particularly when the glass is molten. As

a result, more energy is absorbed by the glass, leading to the formation of a non-uniform molten zone and a subsequent decrease in the final shear force [22]. In addition, the slight displacement observed in the aluminum alloy/glass welds suggests that the joints have poor ductility, which will be discussed in Section 3.4.

Weld microstructure and bonding strength were shown to be correlated in Fig. 6 (b), (c), (d), and (e), respectively. It demonstrated that the glass and aluminum alloy interfaces were winding, with the keyhole and mixing zones forming at the 5052AA side. As can be seen in Fig. 6(b), the molten zone of the 5052AA/glass junction was remarkably confined. Furthermore, the molten area that forms at the 5052AA/weld interfacial region boundary may result in brittle fracture of the weld joint, which would reduce joint strength. On the other hand, as illustrated in Fig. 6(c), the molten zone was significantly extended towards the glass side at the interfacial area. This finding explained the significant differences in bond strength between the examples derived from $E_s=0.336\text{mJ}$ and 0.439mJ , respectively. However, the presence of several defects in the glass/weld interface boundary presents a chance to increase strength at the next pulse energy. Referring to Fig. 6 (d), in the case of $E_s=0.541\text{mJ}$, the fusion zone and interfacial span are remarkably expanded along the 5052AA/weld interface boundary, and the defects in this area were greatly diminished, leading to the development of the bonding shear strength. Although there is little difference in the size of the fusion zone compared to Fig. 6(c) and (d), the numerous pores reduce the welding shear strength, further raising E_s to 0.641mJ .

As stated in Section 3.2.1, the majority of the weld zone was composed of the mixing zone and the modified keyhole zone, which was created during the welding process and subsequently changed. The cross-sectional images demonstrate that variations in the single-pulse energy result in variations in the morphologies of the modified keyhole region and the mixing zone. Fig. 6 (b) shows the initial surface morphology caused by $E_s=0.336\text{mJ}$. Unirradiated patches arise because the laser pulse energy is not enough to drive the momentum of mixing and diffusion at the interfacial region. This could be because of the energy supply limitation, which reduces laser energy absorption because the opaque glass has not formed yet. The result is a severely limited weld zone towards the glass side. However, the aluminum alloy side is locally heated by the lower single-pulse energy, and because the Al atom mass is larger than that of SiO_2 , the Al microparticles readily disperse within the molten zone. However, the restricted energy input may cause incomplete Al movement, which would cause Al microparticles to emerge near the weld boundary.

Up to 0.439mJ of single-pulse energy, as seen in Fig. 6 (c). It is significant to remember that the extra energy additionally raised the molten zone's developing temperature, which caused the glass and aluminum alloy to melt more thoroughly and the mixing zone to widen. The extension mixing may significantly improve the joint's mechanical strength [38]. Consequently, the weld

shear strength increased by 1.5 times compared to the single-pulse energy of 0.439 mJ. Different diffusion behaviours were caused by thermodynamic factors; silicon and aluminum had a weaker affinity for oxygen. Consequently, the suppression of nucleation brought on by the fast cooling rate may result in the generation of the conjunction Al-O at the limited region [21]. Additionally, under the recoil pressure and capillary effect of laser-induced vaporization, aluminum flows readily due to its comparatively low viscosity when molten; in contrast, SiO₂ has a significantly higher viscosity when molten, which makes it more resistant to flow. The keyhole zone forms a heterogeneous morphology as a result of the viscosity difference between silicon dioxide and aluminum.

In Fig. 6(d), the full mixing and diffusion of two interfacial materials encourage the keyhole zone to expand when the single-pulse energy reaches 0.541 mJ. Furthermore, because of the increased temperature and pressure in the middle of the molten zone, the higher laser energy may cause molten aluminum to interact with the glass, producing localized vaporization spots. This resulted in the formation of more crow's foot-shaped keyholes, which, when combined with the extended interfacial span, might significantly increase the 5052AA/glass joints' maximum shear strength.

But when the pulse energy is increased to 0.641 mJ, Fig. 6 (e) shows that there are plenty of inhomogeneous structures in the weld zone in addition to the formation of many micro-holes surrounding the interfacial region. The development of many hot spots in the melt with local temperatures greater than vaporization or even ionization temperatures may be the cause of the creation of huge cavities near the glass/weld zone contact [22].

3.3.2. Effect of laser overlapping factor

Fig. 7 (a) shows the bond strength of the weld under processing conditions $P=10.81\text{ W}$, $f=20\text{ kHz}$ for different laser overlapping factor values, $PO=84\%$, 90% , 94% , and 98% . According to the data, the initial weld strength was $\sigma=12.38\text{ MPa}$. With the augmentation of PO from 84% to 90% , this strength dramatically improved by more than fourfold to 50.08 MPa . However, there was a discernible decrease in σ to 36.60 MPa as PO was raised to 94% . The shear strength was 25.50 MPa as a result of this drop continuing as the PO increased to 98% . Notably, the average bonding strength attained in this investigation exceeded the 12.01 MPa value shown in Ref. [17], where a ns pulsed laser system was also used to directly connect glass and aluminum.

Fig. 7 (a₁) illustrates the force-displacement curves in relation to various pulse overlapping factor values. The shear force of the 5052 aluminum alloy/glass joints show a significant decrease after reaching peak values of 13.73 N , 71.23 N , 54.43 N , and 41.75 N . These peak values correspond to impressive pulse overlapping factors of 84% , 90% , 94% , and 98% , respectively, highlighting the

important relationship between these factors and the quality of bonding. The maximum displacements of the welding zone for the different energy levels were recorded as 0.054mm, 0.219mm, 0.198mm, and 0.177mm. To validate the bonding shear strength values presented in Fig. 7 (a), the mean projected surface area of the welded seam (S) was measured at 1.109mm², 1.422mm², 1.487mm², and 1.637mm², corresponding to the various pulse overlapping factor levels. The maximum shear force in the weld zone notably increases fivefold when the laser overlapping factor rises from 84% to 90%. Enhancing the overlapping factor can significantly boost interdiffusion and chemical reactions, thereby optimizing the microstructures at the metal/glass interface and within the weld [28]. This improvement is crucial for maximizing shear force, leading to stronger and more reliable connections. As the overlapping factor rises to 94% and then to 98%, it is associated with the development of defects in the interfacial region and issues related to non-smooth boundaries. This leads to a significant reduction in the maximum shear force, which drops to 54.43N at 94% and further decreases to 41.75N at 98% PO .

Fig. 7(b), (c), (d), and (e) show the cross-sectional surface morphologies of the weld seam that were formed under various processing conditions and captured using CLSM. Accordingly, the processing settings matched the four scenarios shown in Fig. 7(a). The maximum height of the bonding seam, as established by CLSM, was found to increase with the pulse overlapping factor. The elevation ranged from 24.6 to 37.1μm, and the range of seam dimensions expanded when PO was raised from 84% to 98%. An ns pulsed laser used for glass-to-copper transmission welding, on the other hand, produced a seam height of approximately 59μm with a comparable maximum PO [22]. This discrepancy results from aluminum's lower thermal conductivity, which makes heat transfer less effective during the process and causes less substantial melting of the material at the joint interface. Furthermore, copper's heat-dissipating properties might lead to a better-regulated melting and solidification process, which would encourage the development of a higher seam height in comparison to the 5052AA/SiO₂ joint than aluminum.

The evolution of the seam dimension for Fig. 7(b), (c), and (d) show a convincing microstructure that would support a strong bond strength without any micro-defects at the interfacial region. A greater range of direct laser incident energy absorption around the interfacial region led to the interfacial line lengthening. Concurrently, the elevation rise is ascribed to the increased temperature-induced improvement in the opacity and absorption coefficient along the incoming path of the laser beam. However, Fig. 7 (e) showed an appearance of micropores along the interfacial area and the weld zone's periphery, particularly the glass/weld zone boundary, at $PO=98\%$. Consequently, there was no discernible increase in bond strength in this configuration.

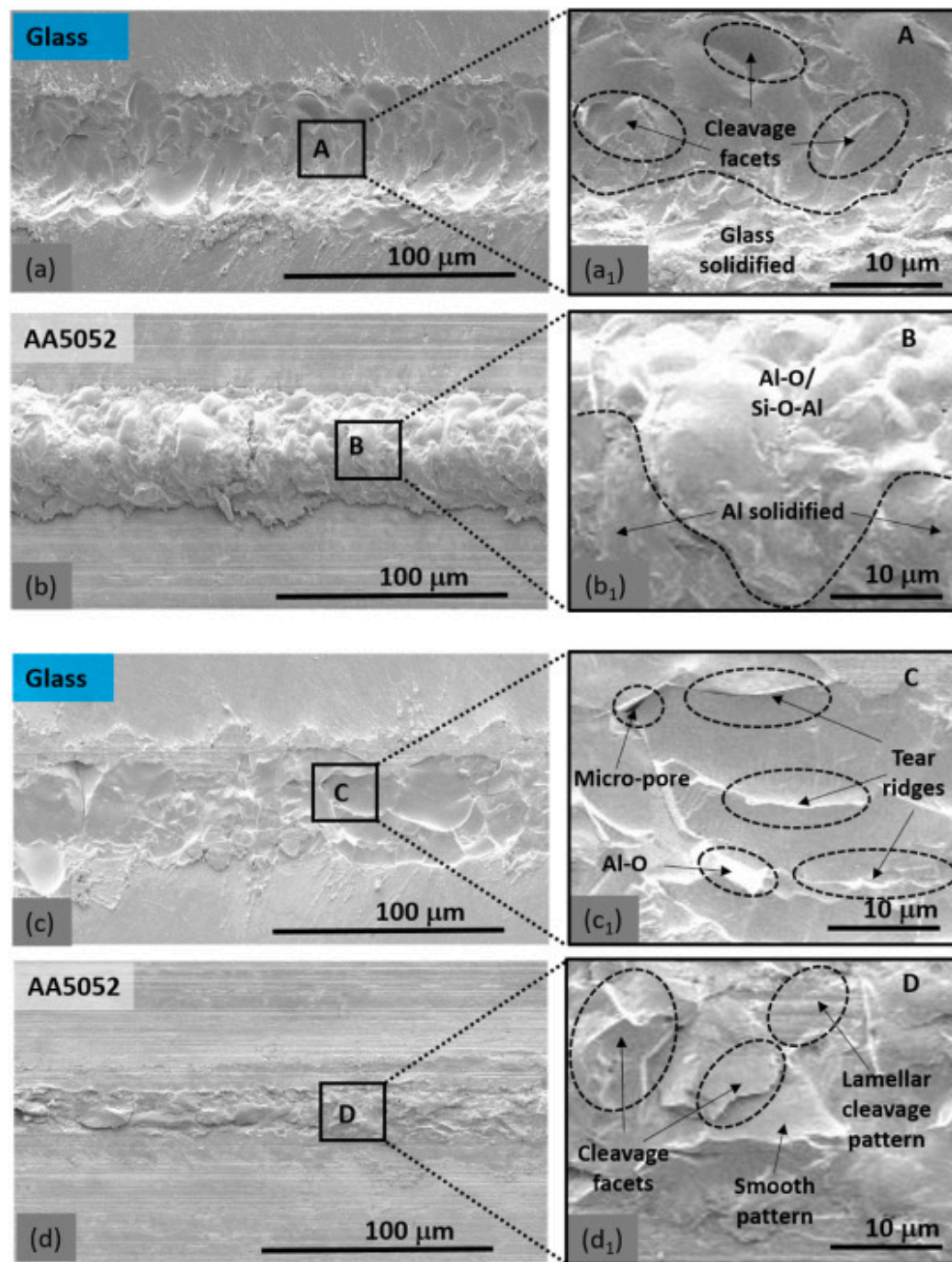
In addition to the different weld seam sizes, the cross-sectional pictures show differing keyhole area morphologies that depend on variations in the pulse overlapping factor. Fig. 7 (b) shows a V-shaped pattern, which changes to a wave-shaped pattern in Fig.

7(c) and (d), and then to a root-like structure in Fig. 7 (e). The laser energy density and energy input are insufficient to provide intense vaporization of the interfacial zone at a lower pulse overlapping factor, which causes the aluminum to melt only along the weld boundary and leaves the development of the keyhole unclear. Furthermore, mixed convection was the predominant mechanism for mixing within the molten zone. This process was fuelled by variations in composition gradients and local pressure differences, which created the molten flow towards the aluminum alloy/weld boundary and helped to form the V-shaped structure shown in Fig. 7 (b).

The input energy increases in proportion to the pulse overlapping factor, which causes total melting and mixing at the interfacial zone. This results in fast vaporization at the aluminum alloy/weld boundary as a vapour-filled cavity is formed when the material is struck by a high-intensity laser beam. The improvement of PO may result in more fluid flow and re-distribution of the molten aluminum. Several ripples with wave-like shapes could occur along the keyhole zone at the same time due to heat accumulation and recurring oscillations in the melt pool, as shown in Fig. 7(c) and (d). As the pulse overlapping factor further increased, the boiling aluminum alloy's vapour pressure forced the molten metal sideways and downward, creating a deep, narrow keyhole zone that helped form the root-like shape. In the meantime, hot spots rise at the glass/weld border as a result of increased absorption along the incoming beam path. As shown in Fig. 7 (e), the development of micro-defects can thereby lower the 5052AA/glass joint's weld quality.

3.4. Fracture surface characteristics

SEM pictures in Fig. 8 (a), (b), (c), and (d) show different surface morphologies on the sides of the glass and aluminum alloy, which correspond to the application of shearing separation force to the welds. Variable laser power with fixed processing settings of $f=20\text{kHz}$, $PO=90\%$, and $v=50\text{mm/s}$. In particular, $P=10.81\text{W}$ for Fig. 8 (a) and (b) results in $E_s=0.541\text{mJ}$, whereas $P=12.81\text{W}$ for Fig. 8 (c) and (d) results in $E_s=0.641\text{mJ}$. To enable careful examination, each image has an enlarged view. As a result, there are four enclosed regions, designated A, B, C, and D for Fig. 8 (a₁), (b₁), (c₁), and (d₁), respectively.



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Fig. 8. SEM images revealing cleaved surfaces following shear force separation. Figures (a) and (b) correspond to bonds produced with various single-pulse energy of $E_s=0.439\text{ mJ}$ ($P=8.79\text{ W}$, $f=20\text{ kHz}$, $PO=90\%$, and $v=50\text{ mm/s}$), $E_s=0.541\text{ mJ}$ ($P=10.81\text{ W}$, $f=20\text{ kHz}$, $PO=90\%$, and $v=50\text{ mm/s}$), respectively. Figures (a₁) and (a₂) depict the surfaces on the separated glass side, while Figures (b₁) and (b₂) show the surfaces on the aluminum alloy side.

When the shear separation force is present, the fracture appears at the weld/glass boundary, giving the aluminum alloy substrates a convex shape (Fig. 8 (b) and (d)) and the glass substrates a concave shape (Fig. 8 (a) and (c)). This is explained by the 5052AA/SiO₂ joint weld zone's interfacial extents. In particular, compared to the weld/aluminum alloy, the weld/glass perimeter was substantially larger. Around the glass side, the separation fracture was more likely to occur due to the larger contact area with the brittle material [22]. High-magnification micrographs illustrating areas A on the glass substrate and B on aluminum alloy substrate in Fig. 8 (a) and (b) are presented in Fig. 8 (a₁) and (b₁), respectively. As depicted in Fig. 8 (a₁), the majority were the perceptible cleavage facets were in the middle area. Secondly, there is a noticeable sign of brittle fracture with hardened glass at the weld's outside edge. On the other hand, region B of Fig. 8 (b₁), which represents the convex aluminum side, has a lighter-coloured area with fracture surface morphology that is primarily composed of Al-O or Si-O-Al systems. Additionally, the grey area close to the field suggests that the remaining area may contain solidified Al under the influence of a single-pulse energy of $E_s=0.541\text{ mJ}$. These results imply that it is unclear whether a fracture is brittle or ductile.

The formation molten zone was heterogeneous as a result of the single-pulse energy being increased to $E_s=0.641\text{ mJ}$, which encouraged uneven compositional changes during the melt re-solidification [22]. A typical fracture surface morphology on the glass side with brittle fracture was shown in Fig. 8 (c₁), as evidenced by the tear ridges' features. Furthermore, a variety of micropores emerged, which negatively impacted the quality of the weld and reduced the bonding strength in Fig. 6 (a). Concurrently, an Al-O junction appeared at a restricted zone due to the increased single-pulse energy; this oxide phase can generate more secondary electrons under SEM detection, making the present region brighter. The identification of glass debris at enhanced single-pulse energy was demonstrated by the development of cleavage facets and lamellar cleavage patterns that alternated with smooth patterns in Fig. 8 (d₁). The lack of Al-O or Si-O-Al components and the development of a lamellar cleavage pattern as a prominent element have an undesirable effect on weld performance and brittle fracture as compared to Fig. 8(b₁) and (d₁).

4. Conclusions

This study used a nanosecond pulsed laser in laser transmission bonding to examine the bonding process between an aluminum alloy plate and an alkali-free glass sheet. Laser power and scanning speed were two processing parameters that were systematically changed. The impact of both the single-pulse energy and the pulse overlapping factor on the final bonding strength was investigated. To understand the weld production mechanisms and their interaction with the bond morphology, microstructure and nanostructure observation, elemental composition, and phase configuration surrounding the weld interfaces and within the weld itself, numerous characterisation approaches were used. These included Confocal Laser Scanning Microscopy (CLSM), scanning electron microscopy (SEM), energy-dispersive spectroscopy (EDS), high-resolution transmission electron microscopy (HR-TEM), and high-resolution X-ray photoelectron spectroscopy (HR-XPS)

The following key conclusions were drawn and compiled from the research:

1. The study simplified and determined the appropriate range of welding parameters, or the window for processing, for our system based on the mechanism of ns laser bonding. A well-formed weld seam between the glass substrate and the aluminum alloy plate, free of any obvious defects, was the standard macrostructure of the aluminum alloy/glass junction. The 5052AA/EAGLE XG® glass joint's microstructure also showed a perfect area with a consistent bonding line.
2. The creation of the modified keyhole zone, the root-like shape, intra-mixing, and inter-particle diffusion of Al and SiO₂ were all crucial to the bonding process, according to the study of SEM pictures and EDS maps. Significantly, the HR-TEM and HR-XPS observations demonstrated the presence of an amorphous Si-O-Al area at the weld interface as well as aluminum oxide, Al₂O₃. This could be essential for improving the quality of the glass and aluminum alloy welding.
3. The maximum bonding strength attained was approximately 50.08MPa, with the ideal parameters being a laser single-pulse energy of 0.541 mJ and a pulse overlapping factor of 90%. This outcome demonstrated the encouraging potential of fabricating glass/aluminum joints for power electronic devices.
4. The growth of the keyhole zone can be facilitated by the full mixing and diffusion of two interfacial materials when the single-pulse energy is increased to the appropriate value. In addition, the high temperature and

pressure at the centre of the molten zone facilitate the presence of the additional keyholes with shapes like crow's feet coupled with the extension interfacial span, resulting in enhancing impressive bonding quality of 5052AA/glass joints. On the other hand, the development of a pulse overlapping factor can lead to the molten aluminum flows and redistribute more freely. Simultaneously, heat accumulation and periodic fluctuations in the melt pool could be formed and create ripples along the keyhole walls with wave-like shapes.

5. Fracture surface analysis showed that at different single-pulse energy values, cleavage facets dominated the fractured glass surface, while the presence of different compositions, including solidified glass, Al–O, and Si–O–Al, significantly influenced the fracture morphology at the aluminum alloy surface. The observation of brittle fracture samples highlighted the influence of laser line energy intensity on the bond morphology after subjecting the samples to shear testing.

CRedit authorship contribution statement

Hoai Nguyen: Writing – review & editing, Writing – original draft, Visualization, Investigation, Funding acquisition, Data curation, Conceptualization. **Jeng-Rong Ho:** Validation, Data curation, Conceptualization.

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Acknowledgments

This research is funded by Funds for Science and Technology Development of the University of Danang under project number B2024-DN06-06.

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